

Aim: Weather forecasting using Random Forest.

CODE:

```
import random
data = [(year, loc, day, hr, round(random.uniform(-20,40),1))
        for year in range(2000,2005)
        for loc in range(10)
        for day in range(1,366)
        for hr in range(24)]
for year in range(2000,2005):
    y_data = [r[4] for r in data if r[0]==year]
    print(f"Year:{year}, Avg:{sum(y_data)/len(y_data):.2f}, Max:{max(y_data)}, Min:{min(y_data)}")
with open('filtered_data.txt','w') as f:
    f.writelines(','.join(map(str,r))+'\n' for r in data if r[4]>30)
```

OUTPUT:

```
= RESTART: C:\Users\ROHIT\Desktop\CS NOTES\Big Data Acquisition
ctical-10.py
Year:2000, Avg:9.95, Max:40.0, Min:-20.0
Year:2001, Avg:10.03, Max:40.0, Min:-20.0
Year:2002, Avg:10.13, Max:40.0, Min:-20.0
Year:2003, Avg:10.02, Max:40.0, Min:-20.0
Year:2004, Avg:10.10, Max:40.0, Min:-20.0
```